

$$\Pi = \Delta S - V \quad \rightarrow \quad \begin{aligned} ds/s &= \mu dt + \sigma dw \\ dB/B &= r dt \end{aligned}$$

$$d\Pi = \Delta \cdot ds - dV$$

$$dV(s,t) = \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} ds + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} (ds)^2$$

$$= \frac{\partial V}{\partial t} dt + \frac{\partial V}{\partial S} ds + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} \sigma^2 S^2 dt$$

$$\stackrel{t \rightarrow t+dt}{\textcircled{\Delta}} = \left( \frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt + \frac{\partial V}{\partial S} (ds) \quad \checkmark$$

$$\textcircled{\Delta} d\Pi = \Delta \cdot ds - dV = \left( \Delta - \frac{\partial V}{\partial S} \right) ds - \left( \frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt$$

$$\Delta = \frac{\partial V}{\partial S}$$

$$r\Pi dt = - \left( \frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) dt$$

$$r \left( \frac{\partial V}{\partial S} S - V \right) = - \left( \frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right)$$

$$\Rightarrow \frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} + rS \frac{\partial V}{\partial S} - rV = 0$$

(BSM PDE)

$$ds/s = (\mu - D)dt + \sigma dw.$$

$$dv =$$

$$\Pi = \Delta \cdot S - V$$

$$d\Pi = \Delta \cdot ds - dv + \underline{\underline{(\Delta S) \cdot D \cdot dt}}$$

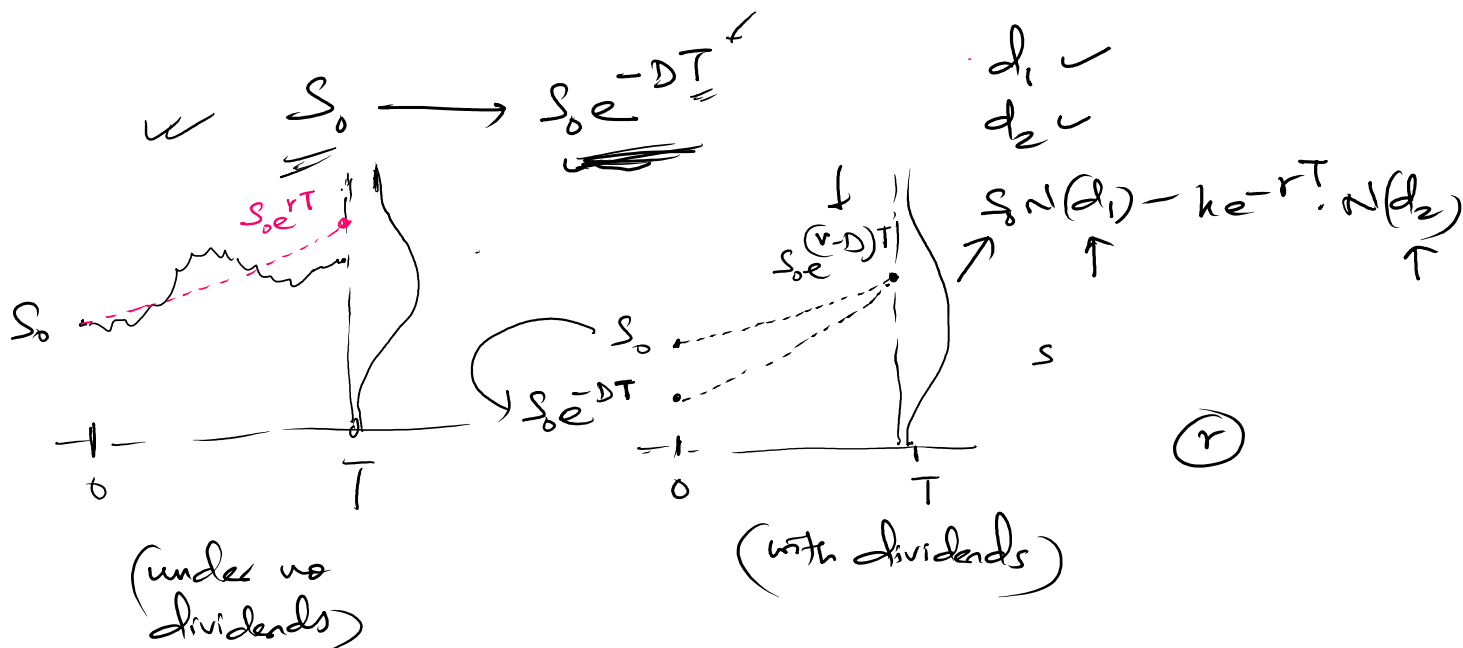
$$dv = \left( \frac{\partial v}{\partial t} + \frac{1}{2} \sigma^2 s^2 \frac{\partial^2 v}{\partial s^2} \right) dt + \frac{\partial v}{\partial s} (ds)$$

$$d\Pi = \left( \Delta - \frac{\partial v}{\partial s} \right) ds - \left( \frac{\partial v}{\partial t} + \frac{1}{2} \sigma^2 s^2 \frac{\partial^2 v}{\partial s^2} - \Delta \cdot s D \right) dt$$

$$r\Pi dt = - \left( \frac{\partial v}{\partial t} + \frac{1}{2} \sigma^2 s^2 \frac{\partial^2 v}{\partial s^2} - \Delta \cdot s D \right) dt$$

$$\Rightarrow r \left( \frac{\partial v}{\partial s} \cdot S - V \right) = 0$$

$$\Rightarrow \frac{\partial v}{\partial t} + \frac{1}{2} \sigma^2 s^2 \frac{\partial^2 v}{\partial s^2} + (r - D) s \frac{\partial v}{\partial s} - rV = 0$$



(under no  
dividends)

(in currencies)

Method. 2

Saturday, August 2, 2025 8:43 AM

$$dB/B = r dt$$

$$\rightarrow V(s, t) = \phi_t \cdot S_t + \psi_t \cdot B_t \quad \leftarrow \text{(self-financing strategy)}$$

$$\underline{dV} = \phi_t \cdot \underline{dS} + \psi_t \cdot \underline{dB} = \phi_t \cdot \underline{dS} + \psi_t \cdot B \cdot r \underline{dt}$$

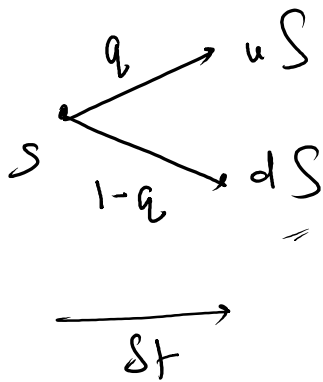
$$\rightarrow \underline{dV} = \left( \frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right) \underline{dt} + \frac{\partial V}{\partial S} \underline{dS}$$

$$\checkmark \Delta = \phi_t = \frac{\partial V}{\partial S}, \quad \underline{\psi_t \cdot B \cdot r} = \left( \frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \right)$$

$$\rightarrow rV = \frac{\partial V}{\partial S} rS + \frac{\partial V}{\partial t} + \frac{1}{2} \sigma^2 S^2 \frac{\partial^2 V}{\partial S^2} \rightarrow \underline{BSM}$$

# Method 3.

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$(q, u, d) \rightarrow 3$  parameters

$$\rightarrow \underline{\underline{S_{t+\delta t} = S_t \cdot e^{(r - \frac{1}{2}\sigma^2)\delta t + \sigma\sqrt{\delta t} \cdot Z}}}$$

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CRP (Cox-Ross-Rubenstein)

$$q = \frac{e^{r\delta t} - d}{u - d}, \quad \underline{u} = e^{\sigma\sqrt{\delta t}}, \quad \underline{d} = e^{-\sigma\sqrt{\delta t}}$$

$ud = 1$  (recombining)

$$S_{t+\delta t} = S_t \cdot e^{(r - \frac{1}{2}\sigma^2)\delta t + \sigma\sqrt{\delta t} \cdot Z}$$

$$X = \ln\left(\frac{S_{t+\delta t}}{S_t}\right) = (r - \frac{1}{2}\sigma^2)\delta t + \sigma\sqrt{\delta t} \cdot Z$$

$$\mathbb{E}(X) = (r - \frac{1}{2}\sigma^2)\delta t, \quad V(X) = \sigma^2\delta t$$

$$\delta t \xrightarrow{f} 0$$

$$e^x = 1 + x + \frac{1}{2}x^2 + \dots$$

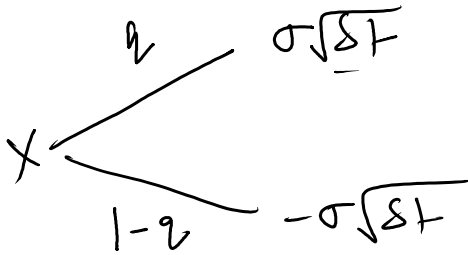
$$S \begin{cases} \nearrow q \cdot S e^{\sigma\sqrt{\delta t}} \\ \searrow 1-q \cdot S e^{-\sigma\sqrt{\delta t}} \end{cases}$$

$$q = \frac{e^{r\delta t} - d}{u - d} = \frac{e^{r\delta t} - e^{-\sigma\sqrt{\delta t}}}{e^{\sigma\sqrt{\delta t}} - e^{-\sigma\sqrt{\delta t}}}$$

$$q \approx \frac{(1 + r\delta t) - (1 - \sigma\sqrt{\delta t} + \frac{1}{2}\sigma^2\delta t)}{(1 + \sigma\sqrt{\delta t} + \frac{1}{2}\sigma^2\delta t) - (1 - \sigma\sqrt{\delta t} + \frac{1}{2}\sigma^2\delta t)}$$

$$q \approx \frac{r\delta t + \sigma\sqrt{\delta t} - \frac{1}{2}\sigma^2\delta t}{2\sigma\sqrt{\delta t}}$$

$$X = \ln \left( \frac{S_{t+\Delta t}}{S_t} \right)$$



$$\underline{S} \rightarrow \underline{S e^{\sigma\sqrt{\Delta t}}}$$

$$\begin{aligned} \mathbb{E}(X) &= q \sigma\sqrt{\Delta t} + (1-q)(-\sigma\sqrt{\Delta t}) \\ &= \underline{2q\sigma\sqrt{\Delta t}} - \underline{\sigma\sqrt{\Delta t}} \\ &= r\Delta t + \cancel{\sigma\sqrt{\Delta t}} - \frac{1}{2}\sigma^2\Delta t - \cancel{\sigma\sqrt{\Delta t}} \\ &= (r - \frac{1}{2}\sigma^2)\Delta t \end{aligned}$$

$$\begin{aligned} V(X) &= q (\underline{\sigma\sqrt{\Delta t}} - (r - \frac{1}{2}\sigma^2)\Delta t)^2 \\ &\quad + (1-q) (-\underline{\sigma\sqrt{\Delta t}} - (r - \frac{1}{2}\sigma^2)\Delta t)^2 \\ &= q (\underline{\sigma^2\Delta t}) + (1-q) (\underline{\sigma^2\Delta t}) \\ &= \underline{\sigma^2\Delta t} \end{aligned}$$

$$\Delta t \rightarrow 0$$

$$V(S, t) \begin{cases} V_u = V(\underline{S e^{\sigma \sqrt{\Delta t}}}, t + \Delta t) \\ V_d = V(S e^{-\sigma \sqrt{\Delta t}}, t + \Delta t) \end{cases}$$

$$f(x + \Delta x) = f(x) + f'(x) \cdot \Delta x + \frac{1}{2} f''(x) (\Delta x)^2 + \dots$$

$$\begin{aligned} V_u &\approx V + \frac{\partial V}{\partial t} (\Delta t) + \frac{\partial V}{\partial S} \cdot \underline{S \Delta t} + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} (S \Delta t)^2 \\ &= V + \frac{\partial V}{\partial t} (\Delta t) + \frac{\partial V}{\partial S} S (e^{\sigma \sqrt{\Delta t}} - 1) + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} S^2 (e^{\sigma \sqrt{\Delta t}} - 1)^2 \end{aligned}$$

$$\begin{aligned} V_u &= V + \frac{\partial V}{\partial t} (\Delta t) + \frac{\partial V}{\partial S} \cdot S (\sigma \sqrt{\Delta t} + \frac{1}{2} \sigma^2 \Delta t) + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} S^2 \sigma^2 \Delta t \\ V_d &= V + \frac{\partial V}{\partial t} \Delta t + \frac{\partial V}{\partial S} \cdot S (-\sigma \sqrt{\Delta t} + \frac{1}{2} \sigma^2 \Delta t) + \frac{1}{2} \frac{\partial^2 V}{\partial S^2} S^2 \sigma^2 \Delta t \end{aligned}$$

$$\Delta t \rightarrow 0$$

$$\Delta = \frac{V_u - V_d}{S_u - S_d} = \frac{2 \cdot S \frac{\partial V}{\partial S} \cdot \sigma \sqrt{\Delta t}}{S (e^{\sigma \sqrt{\Delta t}} - e^{-\sigma \sqrt{\Delta t}})} = \frac{2 \cdot S \cdot \frac{\partial V}{\partial S} \cdot \sigma \sqrt{\Delta t}}{S \cdot 2 \cdot \sigma \sqrt{\Delta t}} = \frac{\partial V}{\partial S}$$

$$\underline{V} = (qV_u + (1-q)V_d) e^{-r\delta t}$$

$$\uparrow = \left( \underset{\uparrow}{q} (\underset{\leftarrow}{V_u} - \underset{=}{V_d}) + \underset{=}{V_d} \right) e^{-r\delta t}$$

=

$$V(s, t) = e^{-r(T-t)} \cdot U(s, t)$$

$$\frac{\partial V}{\partial t} = e^{-r(T-t)} \cdot \frac{\partial U}{\partial t} + U \cdot e^{-r(T-t)} \cdot r$$

$$\frac{\partial V}{\partial s} = e^{-r(T-t)} \cdot \frac{\partial U}{\partial s}$$

$$\frac{\partial^2 V}{\partial s^2} = e^{-r(T-t)} \cdot \frac{\partial^2 U}{\partial s^2}$$

$$e^{-r(T-t)} \cdot \frac{\partial U}{\partial t} + U \cdot r \cdot e^{-r(T-t)} + \frac{1}{2} \sigma^2 s^2 \cdot e^{-r(T-t)} \cdot \frac{\partial^2 U}{\partial s^2} + r s \cdot e^{-r(T-t)} \cdot \frac{\partial U}{\partial s} - r e^{-r(T-t)} \cdot U = 0$$

$$\Rightarrow \frac{\partial U}{\partial t} + \frac{1}{2} \sigma^2 s^2 \frac{\partial^2 U}{\partial s^2} + r s \frac{\partial U}{\partial s} - r U = 0$$